

Faculty of Engineering and Technology
Electrical and Computer Engineering Department

ENCS2340
Digital System

Student name: Omar Ibrahim Badwan
ID:1251487

Digital-project#1

Supervised by **:**
Dr. Mohammad husien

May 2026

Objective

1. **analyze the given Boolean function and understand its logical behavior through systematic evaluation of all possible input combinations**

2. **construct the truth table for the Boolean expression and identify the corresponding output for each input state**

3. **derive the canonical forms of the function, including the Sum of Minterms (SoM) and Product of Maxterms (PoM)**

1 Solutions and Results:

1.1Truth Table

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

1.2 Canonical Forms(SOM and POM)

SoM

F = Σm(1,5,6,7,9,12,13,14,15)

F=w′x′y′z+w′xy′z+w′xyz′+w′xyz+wx′y′z+wxy′z′+wxy′z+wxyz′+wxyz

PoM

F = ΠM(0,2,3,4,8,10,11)

F=(w+x+y+z)(w+x+y′+z)(w+x+y′+z′)(w+x′+y+z)(w′+x+y+z)(w′+x+y′+z)(w′+x+y′+z′)

1.3 Boolean Simplification Steps:

F(w,x,y,z)=xy′z+x′y′z+w′xy+wxy′+wxy

Complement Law:

F(w,x,y,z)=y′z+w′xy+wxy′+wxy

Complement Law again :

F(w,x,y,z)=y′z+w′xy+wx

Factoring:

F(w,x,y,z)=y′z+x(w′y+w)

Absorption Law:

F(w,x,y,z)=y′z+x(y+w)

Distributive Law:

F(w,x,y,z)=y′z+xy+xw

Final result:

F(w,x,y,z) = xw+ xy + y′z

1.4 Circuit Sehematics and Simulations
1.4.1 Original Function
Circuit Sehematics :

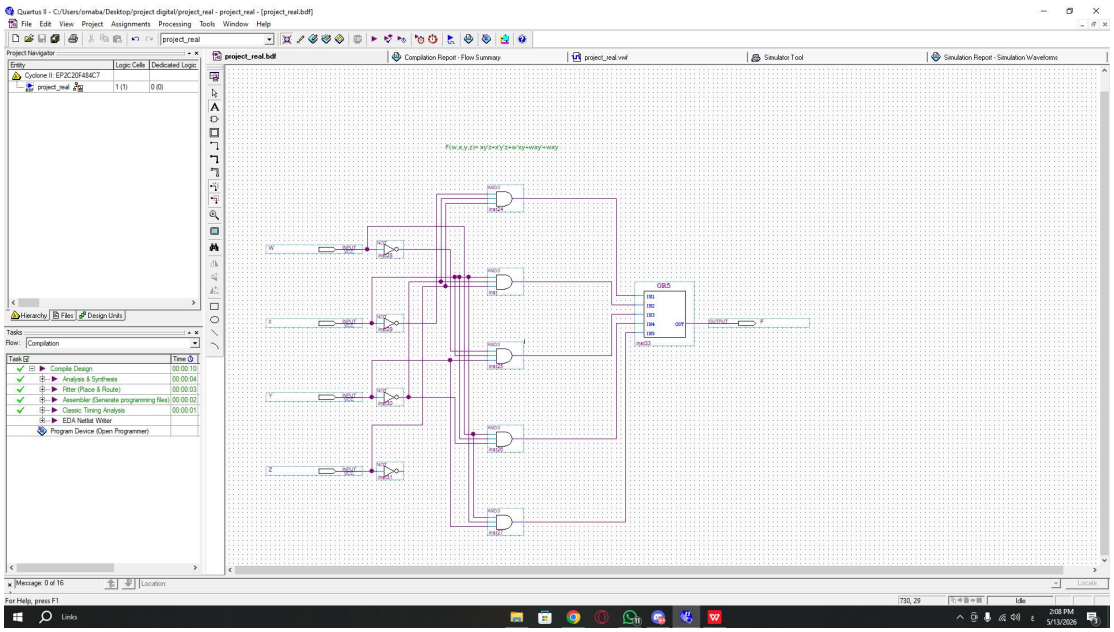


Figure 1: Circuit Sehematics For Original Function

Simulations:

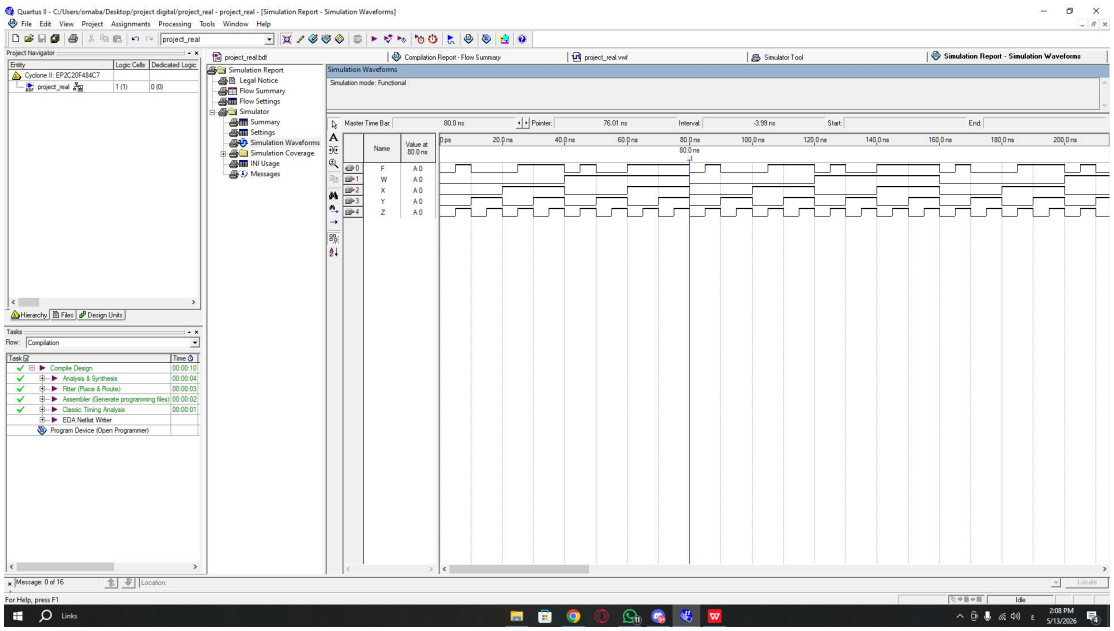


Figure 2: Simulations For Original Function

1.4.2 Minimization Function

Circuit Sehematics :

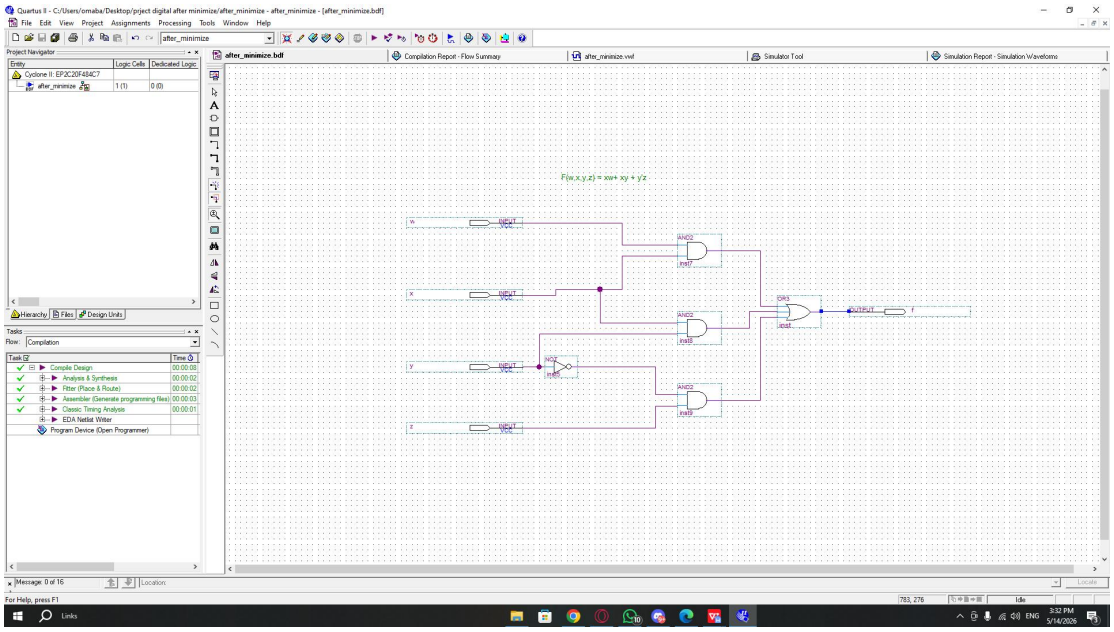


Figure 3: Circuit Sehematics For Minimized Function

Simulations:

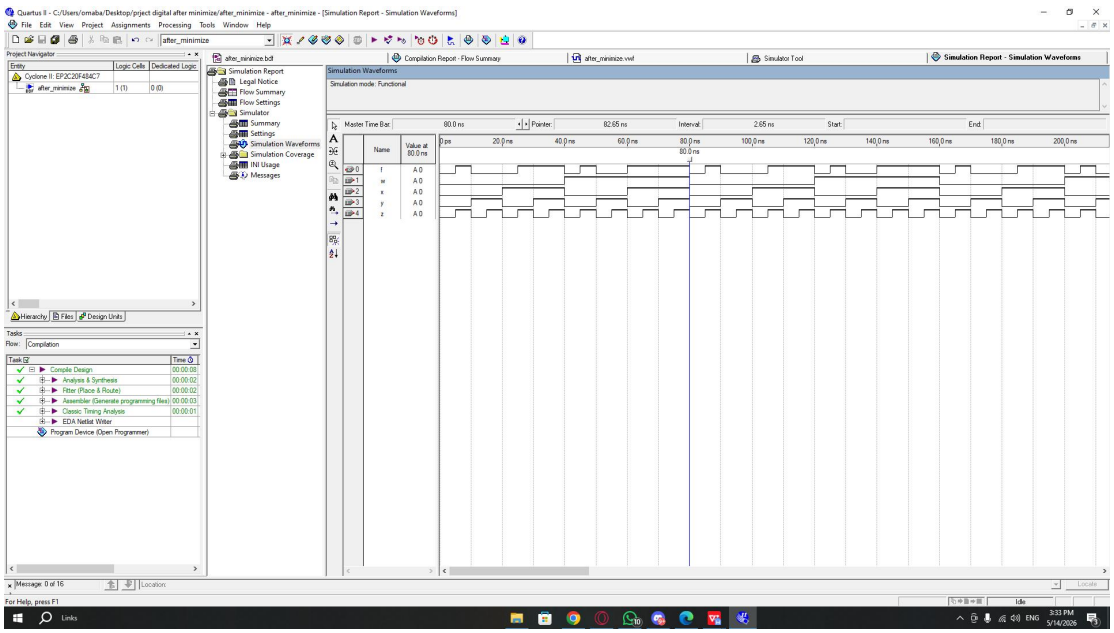


Figure 4: Simulations For Minimized Function

2 Conclusion:

In this project , the given function was analyzed by obtaining the truth table and the canonical forms (SoM) and (PoM) .

I implemented this function using logic gates in Quartus , and then then simplified it using boolean algebra laws , and fortunally both were simulated successfully.

The results obtaiend were correct , and the simulation output matched the tuth table
For all possible inputs , this can confirme the accuracy of both the original and the simplified function.

And what we learned after simplifying the boolean function , it became caler that minimized the circuit required fewer logic gates and a simpler design compared to the original circuit.